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AI for Bharat Hackathon

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Team Name : Bharat Brain Wave

Team Leader Name : Ritesh Raut

BHARAT BRAINWAVE

Problem Statement : AI for Retail, Commerce & Market Intelligence:

Build an AI-powered solution that enhances decision-making, efficiency, or user experience across retail, commerce, and marketplace ecosystems.

Brief about the Idea:

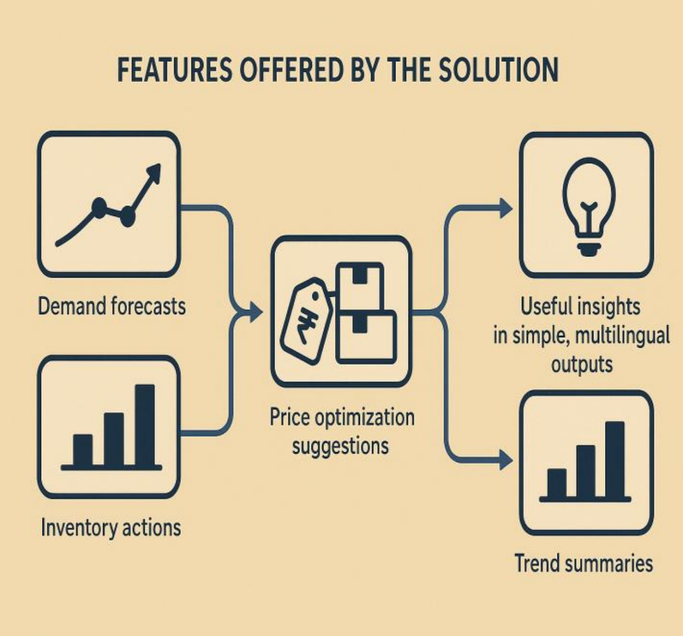
Solution Title: Merchant Intelligence Copilot for Indian MSMEs

Merchant Intelligence Copilot is an AI-powered decision assistant for Indian MSMEs and sellers.

It converts simple CSV / POS data into:

- Demand forecasts
- Price optimization suggestions
- Inventory actions
- Trend summaries
- Useful insights in simple, multilingual outputs

Built using Amazon Bedrock, it delivers actionable recommendations, helping sellers reduce stockouts, avoid overstocking, improve margins, and adopt data-driven planning without complexity.



Explanation of the Solution

How is it different from existing ideas?

- Designed specifically for Bharat MSMEs with low-data onboarding
- Multilingual insights (English/Hindi/Marathi)
- Explains “why” behind suggestions, not just graphs
- Combines forecasting + LLM reasoning + actionable next steps

How will it solve the problem?

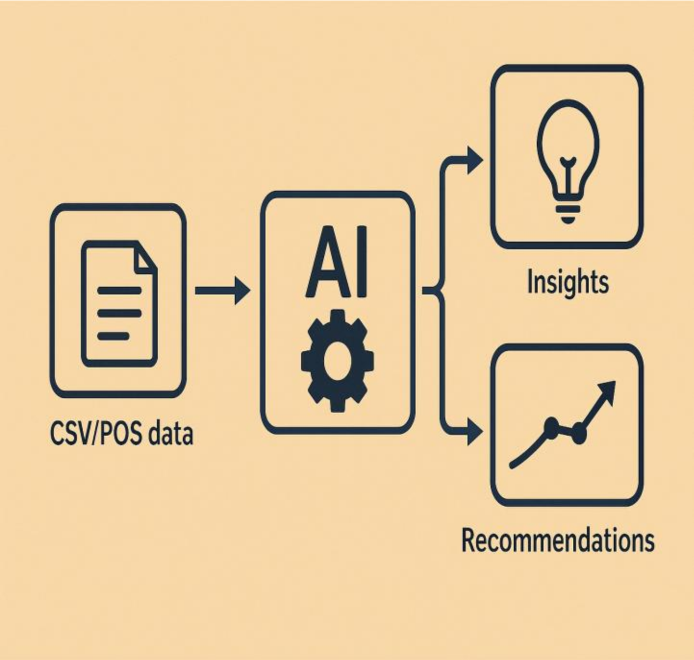
- Forecast weekly/daily product demand
- Suggest reorder quantities and highlight risks
- Recommend optimal price bands based on market signals and margins
- Generate simple action plans sellers can execute immediately

USP (Unique Selling Proposition)

- Actionable Copilot, not just dashboards
- Works with simple CSVs, no specialized software required
- High interpretability with confidence scoring
- Lightweight deployment — web app / API today, WhatsApp-style later

List of features offered by the solution

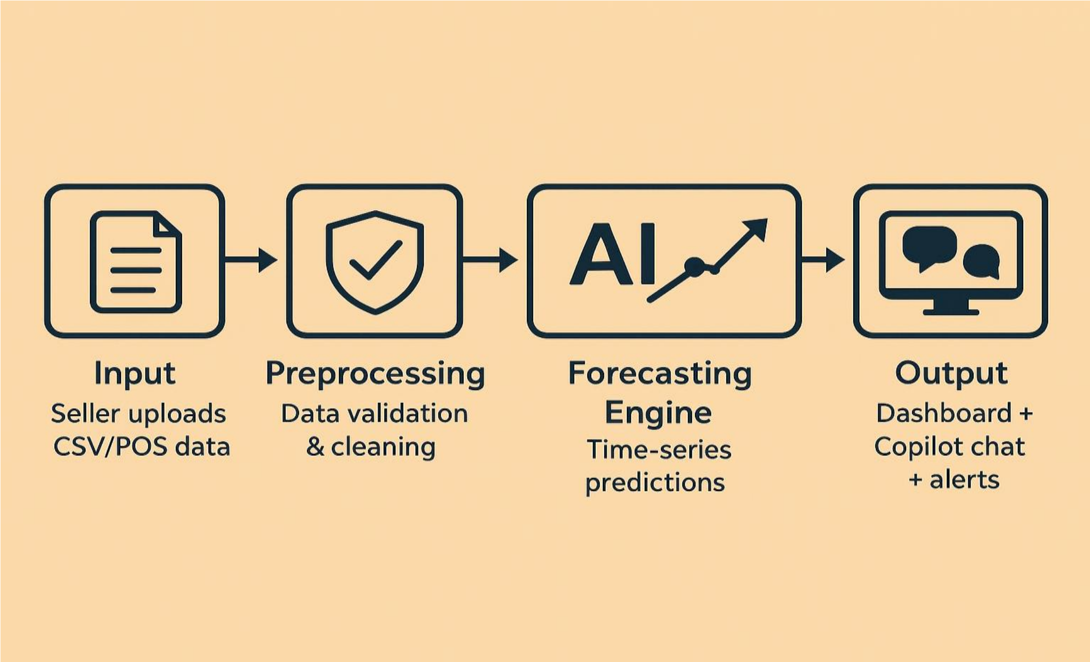
- Upload CSV/POS data (sales, inventory, pricing)
- Automated demand forecasting using seasonal patterns
- Reorder quantity recommendations
- Multilingual insights & explanations
- Price intelligence (margin-aware ranges)
- Trend summarizer (identify spikes/drops)
- Merchant Copilot Chat (Q&A + explanations)
- Alerts for stockout risk, slow movers, anomalies
- Auto-generated weekly action plans
- Shareable reports for store teams



Process flow diagram or Use-case diagram

Process Flow:

- Input: Seller uploads CSV/POS data
- Preprocessing: Data validation & cleaning
- Forecasting Engine: Time-series predictions
- LLM Reasoning: Insights + “why” + action steps
- Output: Dashboard + Copilot chat + alerts



Primary Use Cases:

- “What should I stock for next week?”
- “Which items are slow movers and why?”
- “Suggest optimal price for Product X.”
- “Give me a weekly action plan.”

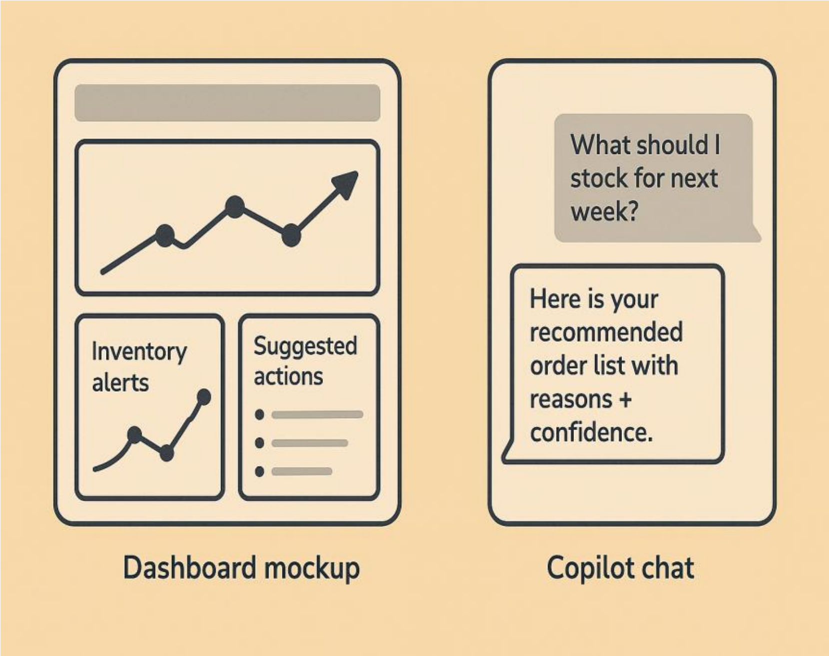
Wireframes/Mock diagrams of the proposed solution

Dashboard Mockup Includes:

- Data upload section
- Sales + forecast chart
- Inventory alerts
- Suggested actions

Copilot Chat Mockup Includes:

- User: “What should I stock for next week?”
- AI: “Here is your recommended order list with reasons + confidence.”



Architecture diagram of the proposed solution:

Components:

Web/Mobile UI: Data upload, dashboard, insights

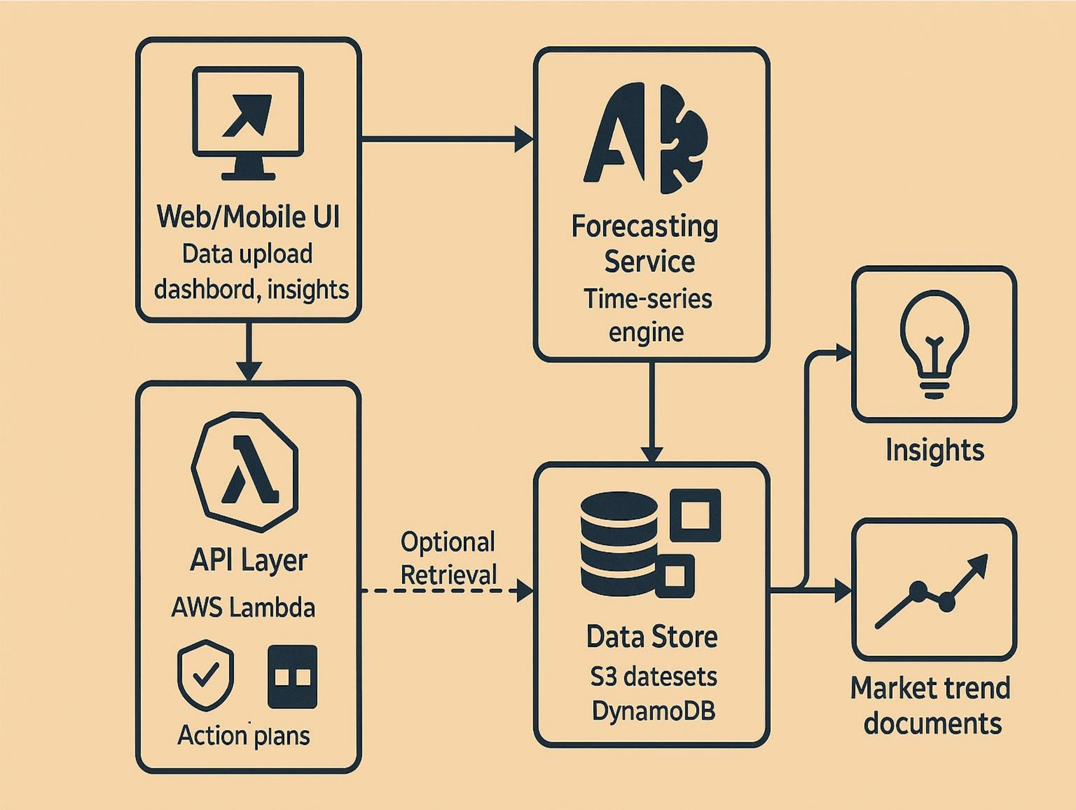
API Layer: AWS Lambda + API Gateway

Forecasting Service: Time-series engine

Amazon Bedrock: LLM for summaries, reasoning, action plans

Data Store: S3 for datasets, DynamoDB for metadata

Optional Retrieval: Market trend documents



Technologies to be used in the solution:

- **Amazon Bedrock –** LLM reasoning, insights, summarization
- **AWS Lambda + API Gateway –** Backend services
- **Amazon S3 –** Data storage
- **DynamoDB –** Metadata storage
- **Forecasting –** Python time-series model
- **Frontend –** React / Next.js or Streamlit for prototype
- **CI/CD –** GitHub
- **Responsible AI:** Input filtering, disclaimers, confidence scoring

Estimated implementation cost :

Prototype Assumptions:

1 merchant

100 LLM queries/day

Small CSV uploads

Cost Optimizations:

Cache repeated insights

Use smaller models for routine tasks

Limit context window

Estimated Monthly Cost:

- Bedrock inference: ₹1,500 – ₹6,000
- S3 + Lambda + API Gateway: ₹300 – ₹1,200

Total: ₹1,800 – ₹7,200 / month

Additional requirements or checklists for the hackathon:

Submission Readiness:

- Meaningful AI usage (forecasting + LLM reasoning)
- Clear "Why AI?" explanation
- Simple & usable UI
- Responsible AI: confidence, safety, disclaimers
- Deliverables planned:

– GitHub repo

– 3-min demo video

– AWS Builder Center blog

– Submission deck

Success Metrics:

- Reduce stockouts by 10–20%
- Reduce slow-moving inventory by 5–15%
- Improve margins by 1–3%

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Thank You

Team: Bharat Brain Wave | **Lead:** Ritesh Raut | **Solution:** Merchant Intelligence Copilot